

Route G, Scheme A: One Internal Spectrum and Physical Probes

Research record: G0–G1

22 September 2026

Abstract

We construct a bounded, falsifiable realization of the common-spectrum idea: a positive-energy complex scalar on a fixed circle, coupled to two specified probes. One action determines both the mass tower and its visibility. A passive basis change cannot change masses; a different physical background can change them. Degenerate standing waves reveal an important limitation: even two probes need not observe every state. This is a conditional field-theory prototype, not a derivation of elementary-particle masses, a string completion, or the full claims of the motivating energy-spectrum document.

1 Declared action, parameters and scope

Use natural units and signature $(+ - - -)$ on $\mathbb{M}^{3,1} \times S^1$:

$$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu - R^2 d\theta^2, \quad \theta \sim \theta + 2\pi, \quad R > 0. \quad (1)$$

The periodic complex scalar has an external flat connection $A_\theta = \alpha$, $D_\theta = \partial_\theta + i\alpha$. The gauge-invariant holonomy is $\exp(2\pi i\alpha)$; α is defined modulo integers. The action is

$$S_0 = \int d^4x \int_0^{2\pi} R d\theta \left(\partial_\mu \Phi^\dagger \partial^\mu \Phi - \frac{|D_\theta \Phi|^2}{R^2} - M_5^2 |\Phi|^2 \right), \quad M_5^2 \geq 0. \quad (2)$$

There is one common R, M_5, α , not one parameter set per particle. Their dimensions are $[R] = -1$, $[M_5] = 1$, $[\alpha] = 0$ and $[\Phi] = 3/2$. The connection and metric are prescribed backgrounds: no dynamical gauge field, gravity, radion stabilization or preferred holonomy is supplied. These omissions limit physical interpretation; they do not invalidate a calculation conditional on (2).

This is an additional *spatial* dimension of a specified model, not a reinterpretation of a Spin(10) representation dimension. The model does not identify its variables with the source document's E_i, E_p, E_n , its symbol R , or negative Hamiltonian energies. It retains a common-spectrum intuition from that document [1], using the conventional compactification mechanism [2, 3]. Route F is unchanged.

2 Canonical reduction and positive energies

The orthonormal eigenfunctions of the internal operator are

$$u_n(\theta) = \frac{e^{in\theta}}{\sqrt{2\pi R}}, \quad \int_0^{2\pi} R d\theta u_m^* u_n = \delta_{mn}, \quad -D_\theta^2 u_n = (n + \alpha)^2 u_n. \quad (3)$$

Expanding $\Phi = \sum_{n \in \mathbb{Z}} \phi_n(x) u_n(\theta)$ in (2), orthogonality gives the canonically normalized four-dimensional action

$$S_0 = \sum_n \int d^4x (|\partial_\mu \phi_n|^2 - m_n^2 |\phi_n|^2), \quad \boxed{m_n^2 = M_5^2 + \frac{(n + \alpha)^2}{R^2}}. \quad (4)$$

Here $[\phi_n] = 1$. With sources switched off, the classical Hamiltonian is

$$H = \sum_n \int d^3x \left(|\dot{\phi}_n|^2 + |\nabla \phi_n|^2 + m_n^2 |\phi_n|^2 \right) \geq 0. \quad (5)$$

After quantization and the usual vacuum subtraction, particle and antiparticle excitations have positive $\omega_n(\mathbf{p}) = \sqrt{\mathbf{p}^2 + m_n^2}$. A negative Fourier label n denotes compact momentum, not negative energy. Opposite frequency solutions do not introduce negative-norm states. The massless special case $M_5 = 0$, $n + \alpha = 0$ requires the usual care at zero external momentum.

For adjacent *signed Fourier labels*, not adjacent sorted masses,

$$m_{n+1}^2 - 2m_n^2 + m_{n-1}^2 = \frac{2}{R^2}, \quad \Delta^3 m_n^2 = 0. \quad (6)$$

This is a restrictive, testable tower relation. Four or more assigned consecutive branches cannot have arbitrary independent masses. A large light-to-heavy hierarchy is possible near a massless branch; there is no claim that the circle forbids every hierarchy.

3 Visibility derived from a fixed interaction

Place probes at $0, \pi$ and choose the positive arc from 0 to π . Parallel transport to the first probe is

$$W(0, \pi) = \exp \left(i \int_0^\pi A_\theta d\theta \right) = e^{i\pi\alpha}, \quad \mathcal{O}_\pm = \frac{\Phi(0) \pm W(0, \pi)\Phi(\pi)}{\sqrt{2}}. \quad (7)$$

Specify one coupling, with no mode-dependent projector choices:

$$S_{\text{probe}} = \int d^4x \left[\lambda J_+^\dagger \mathcal{O}_+ + \lambda J_-^\dagger \mathcal{O}_- + \text{h.c.} \right]. \quad (8)$$

We take $[\lambda] = 0$, $[J_\pm] = 5/2$, $\lambda \neq 0$. The sources are external; this is linear response, not a dynamical detector or a particle-conversion process.

Under $\Phi \mapsto e^{i\chi} \Phi$ and $A_\theta \mapsto A_\theta - \partial_\theta \chi$,

$$W \mapsto e^{-i(\chi(\pi) - \chi(0))} W, \quad \mathcal{O}_\pm \mapsto e^{i\chi(0)} \mathcal{O}_\pm. \quad (9)$$

The operators are *charged and gauge-covariant*, not local gauge-invariant observables of a dynamical gauge theory. Transforming $J_\pm$ with the same charge makes (8) invariant. Changing the chosen path in a nontrivial holonomy changes the physical probe definition, not merely a gauge convention.

Writing $\lambda \mathcal{O}_a = \sum_n g_{an} \phi_n$ gives

$$g_{\pm n} = \frac{\lambda}{\sqrt{4\pi R}} (1 \pm e^{i\pi(n+\alpha)}), \quad (10)$$

$$\rho_{ab}^{(n)} = g_{an} g_{bn}^*, \quad \rho^{(n)} \succeq 0, \quad (11)$$

$$|g_{\pm n}|^2 = \frac{|\lambda|^2}{2\pi R} [1 \pm \cos \pi(n + \alpha)], \quad (12)$$

$$\rho_{+-}^{(n)} = \frac{i|\lambda|^2}{2\pi R} \sin \pi(n + \alpha), \quad (13)$$

$$\text{tr } \rho^{(n)} = \frac{|\lambda|^2}{\pi R}. \quad (14)$$

The normalized weights $p_{\pm n} = (1 \pm \cos \pi(n + \alpha))/2$ sum to one, but are *relative probe strengths*, not branching ratios.

For $Q^2 \geq 0$, away from a massless zero-momentum singularity, the Euclidean source response is

$$\mathcal{G}_{ab}(Q^2) = \sum_{n \in \mathbb{Z}} \frac{\rho_{ab}^{(n)}}{Q^2 + m_n^2}. \quad (15)$$

Its continuation has the same possible poles in every channel; a zero residue can hide a pole from a particular channel. For every source vector v , $v^\dagger \mathcal{G} v = \sum_n |g_n^\dagger v|^2 / (Q^2 + m_n^2) \geq 0$. The response has dimension -1 in these conventions.

Degeneracy: a genuine dark subspace

For a mass eigenvalue s , let C_s have columns g_n for all n with $m_n^2 = s$. The invariant pole residue is

$$R_s = C_s C_s^\dagger, \quad \dim \ker C_s = d_s - \text{rank } C_s, \quad \text{tr } R_s = d_s \frac{|\lambda|^2}{\pi R} > 0. \quad (16)$$

The branch sum (14) holds in the Fourier basis; it is *not* a claim of equal strength for every superposition in a degenerate eigenspace. At $\alpha = 0$, the $n = k, -k$ columns coincide for $k \neq 0$. Their antisymmetric standing wave

$$\frac{u_k - u_{-k}}{\sqrt{2}} = \frac{i \sin(k\theta)}{\sqrt{\pi R}} \quad (17)$$

vanishes at both probe positions. Thus $\text{rank } C_s = 1$ and a massive state can be dark to *both* probes. The orthogonal combination remains visible: the entire mass pole does not disappear from both channels. At $\alpha = 1/2$, the degenerate pair $n, -n - 1$ has independent (indeed orthogonal) coupling columns, so this two-dimensional dark subspace obstruction is absent. This is an apparatus-dependent result, not a universal rule of nature.

4 What does and does not change a mass

In any retained mode space, let $K = Q^2 I + M^2$ and let B have columns g_n . Under $\phi = U \tilde{\phi}$ with U unitary,

$$\tilde{K} = U^\dagger K U, \quad \tilde{B} = B U, \quad \tilde{B} \tilde{K}^{-1} \tilde{B}^\dagger = B K^{-1} B^\dagger. \quad (18)$$

The masses and the transformed response do not change. Rotating only a mass matrix or only a probe is not a passive change of the same experiment. For a general invertible field redefinition the kinetic metric must also be transformed.

An integer large gauge transformation is

$$\alpha \mapsto \alpha + \ell, \quad \Phi \mapsto e^{-i\ell\theta} \Phi, \quad n \mapsto n - \ell. \quad (19)$$

Both $n + \alpha$ and the probe coefficients remain unchanged. A finite cutoff comparison must relabel the retained window too. By contrast, physically changing R or a gauge-inequivalent holonomy can change the spectrum. Not every distinct value of α gives a distinct mass set: α and $-\alpha$ have the same mass set after reflection of labels. A comparison of two static backgrounds does not describe the energy cost, vacuum selection, or time evolution of such a change.

5 Numerical verification and truncation control

No experimental masses are input. The four engineering cards have $M_5 = 0.5$ and $\lambda = 1$ in common arbitrary energy units. For the branch $n = 0$:

R	α	m_0^2	p_{+0}	p_{-0}
1	0	0.250000	1	0
1	0.25	0.312500	0.853553	0.146447
1	0.5	0.500000	0.5	0.5
2	0.25	0.265625	0.853553	0.146447

The decimals display exact formulae, not fitted coefficients. At $\alpha = 0, n = 1, m_1^2 = 1.25$ but $p_{+1} = 0$: mass and visibility are different observables.

An independent matrix discretization uses $h = 2\pi/N$, $\theta_j = jh$, $U_j = e^{i\alpha h}$ and periodic indices:

$$(L_h\psi)_j = M_5^2\psi_j + \frac{2\psi_j - U_j\psi_{j+1} - U_{j-1}^*\psi_{j-1}}{R^2h^2}. \quad (20)$$

Its quadratic form is $Rh \sum_j [M_5^2|\psi_j|^2 + |U_j\psi_{j+1} - \psi_j|^2/(R^2h^2)] \geq 0$. The matrix eigenvalues are checked against

$$m_{n,h}^2 = M_5^2 + \frac{4}{R^2h^2} \sin^2 \left(\frac{(n+\alpha)h}{2} \right) = m_n^2 - \frac{(n+\alpha)^4h^2}{12R^2} + O(h^4). \quad (21)$$

Only resolved low labels are used for continuum convergence; the lattice's finite Brillouin zone is not an extra prediction. For a small gauge change, $U_j \mapsto e^{i\chi_j}U_j e^{-i\chi_{j+1}}$, transform both the matrix and the discrete Wilson-dressed probes. Degenerate comparisons use (16), not individual numerical eigenvectors.

For the symmetric mode truncation $|n| \leq N_{\text{mode}} > |\alpha|$, every response matrix entry obeys the conservative bound

$$|\mathcal{G}_{ab} - \mathcal{G}_{ab}^{(N_{\text{mode}})}| \leq \frac{|\lambda|^2 R}{\pi} \sum_{|n| > N_{\text{mode}}} \frac{1}{(|n| - |\alpha|)^2} \quad (22)$$

$$\leq \left\lfloor \frac{2|\lambda|^2 R}{\pi(N_{\text{mode}} - |\alpha|)} \right\rfloor. \quad (23)$$

This follows from $|\rho_{ab}^{(n)}| \leq |\lambda|^2/(\pi R)$, $Q^2, M_5^2 \geq 0$ and the integral bound for a decreasing positive series. A difference between two finite cutoffs is not itself the exact infinite-tail error. The implementation records both successive differences and this a priori bound.

An independent infinite-sum check follows from Poisson summation. For $\beta = R\sqrt{M_5^2 + Q^2} > 0$, Fourier transforming $(q^2 + \beta^2)^{-1}$ yields

$$F(x) = \sum_n \frac{e^{i(n+\alpha)x}}{(n+\alpha)^2 + \beta^2} = \frac{\pi}{\beta} \sum_{\ell \in \mathbb{Z}} e^{2\pi i \ell \alpha} e^{-\beta|x-2\pi\ell|}. \quad (24)$$

Set $a = e^{-\pi\beta}$, $t = 2\pi\alpha$ and $D = 1 + a^4 - 2a^2 \cos t$. Geometric summation at $x = 0, \pi$ gives

$$S_0 = F(0) = \frac{\pi}{\beta} \frac{1 - a^4}{D}, \quad S_\pi = F(\pi) = \frac{\pi}{\beta} \frac{a(1 - a^2)(1 + e^{it})}{D}, \quad (25)$$

and therefore

$$\mathcal{G} = \frac{|\lambda|^2 R}{2\pi} \begin{pmatrix} S_0 + \text{Re } S_\pi & i \text{Im } S_\pi \\ -i \text{Im } S_\pi & S_0 - \text{Re } S_\pi \end{pmatrix}. \quad (26)$$

All four cards have $\beta > 0$ at the tested momenta. This formula provides an exact response reference, not a large-cutoff surrogate.

The executable and machine-readable report are respectively

```
route_g/code/verify_g1_circle_spectrum.py
route_g/output/g1_circle_spectrum.json.
```

They record matrix convergence, full residues, gauge/basis covariance, dark-subspace ranks, positivity, truncations and source hash. Checks establish implementation consistency, not experimental evidence.

6 Next physical gate: an actual transition

The free action is diagonal in n . It supplies no spontaneous $\phi_n \rightarrow \phi_m$ conversion. A prescribed external source may inject energy and excite a mode, but is not yet an energy-accounted dynamical detector.

Even a natural bulk extension can fail before any lengthy calculation. Consider a massless periodic neutral scalar χ and the candidate vertex $-\lambda_5 \chi |\Phi|^2$ (not a completed stable potential). Its Fourier overlap enforces $n = m + q$, and $m_{\chi_q} = |q|/R$. For $M_5 > 0$ and $n \neq m$, the strict Euclidean triangle inequality for $(M_5, (m + \alpha)/R)$ and $(0, (n - m)/R)$ gives

$$\sqrt{M_5^2 + \frac{(n + \alpha)^2}{R^2}} < \sqrt{M_5^2 + \frac{(m + \alpha)^2}{R^2}} + \frac{|n - m|}{R}. \quad (27)$$

Hence $\phi_n \rightarrow \phi_m + \chi_q$ is kinematically closed. For $M_5 = 0$ the non-strict inequality still holds; equality supplies only threshold kinematics, not an ordinary open two-body decay phase space. Other processes, scattering, extra fields or modified backgrounds are not excluded by this particular argument.

G2 should therefore choose *one* localized dynamical interaction/detector, or one explicitly energy-accounted driven background, and compute one allowed process. Localization is a new physical assumption: compact momentum must be exchanged with that structure, not silently violated. G3 may study background selection; G4 must derive spin, charges and chirality before a particle identification or a Route-F bridge is proposed.

Bounded conclusion. G1 realizes one common positive spectrum with interaction-defined visibility and exact correlated tower relations. It rejects coordinate-induced mass conversion, while allowing genuine dark standing waves for an incomplete apparatus. It does not repair Route-F flavor/seesaw tensions, derive three chiral families, or prove the motivating document. The next useful step is a transition amplitude with a stated source of energy and momentum, not another mass fit.

References

- [1] B. Chen, *Energy spectrum for elementary particles*, final conference document (2022), five pages. Exact final PDF. Source checksum and interpretation ledger: `route_g/SOURCES.md`.
- [2] D. Tong, *Lectures on String Theory*, section 8, arXiv:0908.0333. No string oscillator or winding sector is assumed here.
- [3] A. Yamada, *The UV sensitivity of the Higgs potential in Gauge-Higgs Unification*, PTEP 2021, 093B01, section 2, equation (10), doi:10.1093/ptep/ptab085.